

Buffer Overflow Attacks

2. Running ./main 6 returns:

Target Function Ptr: 0x4011dc

3.

Running ./main 8 gives the same output, but when an offset of 50 is used we get a segmentation fault error.

Checking more carefully, the offset threshold for which we get a 'segmentation fault (core dumped)' is at 14.

At an offset of 20, the error changes to 'bus error (core dumped)', however this only occurs for offsets 20 and 21, and all other offsets above 14 give the segmentation error, not the bus error.

At offset 24, we see that the program returns, 'Success! Malicious Function Called', suggesting that the buffer address was overwritten with a pointer to the malicious function for this offset.

The target function pointer for all runs of the function with all offsets is always 0x4011dc

4.